One model to test sequence redesign

Train contacts-v1 Qwen3 1.5B from scratch on native and ProteinMPNN-redesigned AFDB and ESM-Atlas documents. Companion to Zack's broader issue #274 sweep; tracked in issue #277.

Fixed first-pass recipe

Use exp232's winning m2-p06 optimizer: LR 0.001, weight decay 0.2. Preserve 5.9522% AFDB / 94.0478% ESM sampling and split each source equally between native and redesigned documents. Train for 145,200 steps (152.253B tokens), retaining amino-acid order augmentation and the original WSD schedule.

Placement and current status

Iris inventory identified US-EAST-02A as the regional home of all four inputs. At placement time, 208 H100s were unallocated across 26 nodes. Plan one 16-node / 128-H100 batch-priority gang, after CPU tokenization and an isolated training smoke. No model-quality result is available yet.

Interpretation

Compare first with exp232's matched-budget native-only checkpoint. The longer-trained best model is a separate reference. One seed and one mixture can provide an early answer but cannot settle the broader mixture sweep.